

LECTURE

WHAT IS A VIOLIN PLOT?

SECTION: CREATE A VIOLIN PLOT ON CHATGPT & JULIUS AI



IN THIS LECTURE, YOU WILL LEARN

WHAT IS A VIOLIN PLOT?

WHY IS IT IMPORTANT?

HOW IS THE VIOLIN PLOT USED IN SIX SIGMA PROJECTS?

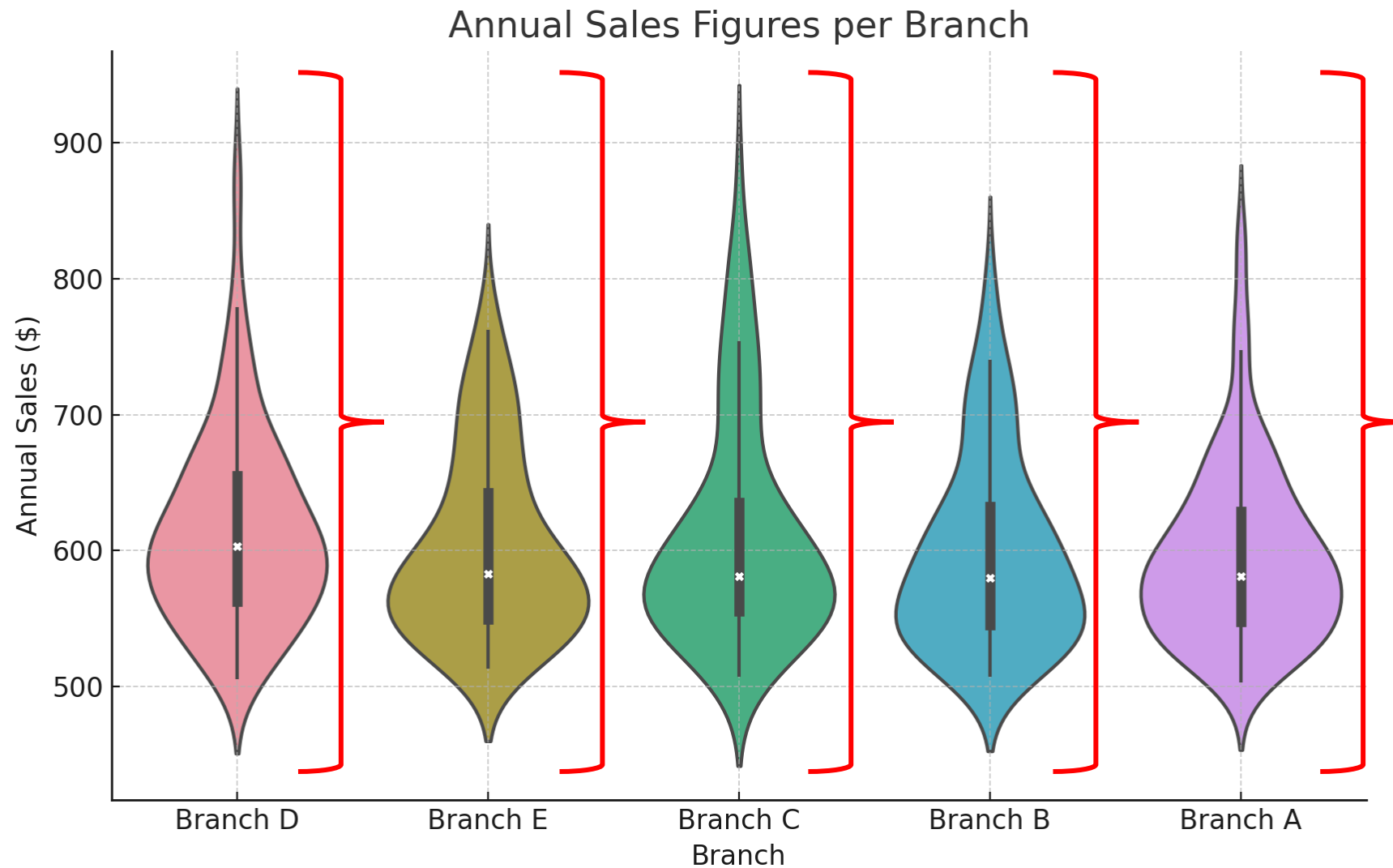
WHAT ARE THE COMPONENTS OF A VIOLIN PLOT?

UNDERSTANDING THE DATA IN A VIOLIN PLOT

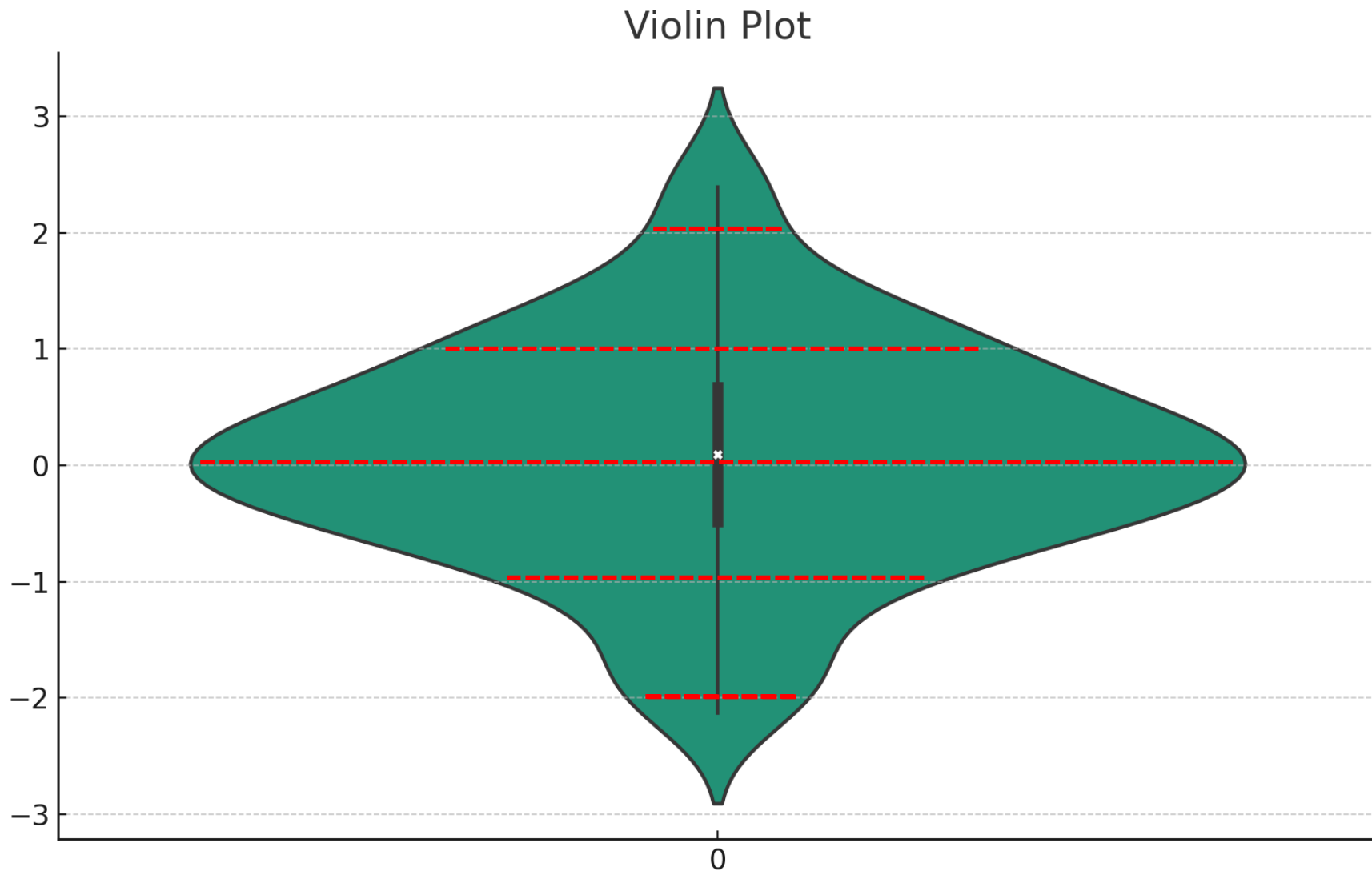
CHAT GPT

WHAT IS A VIOLIN PLOT?

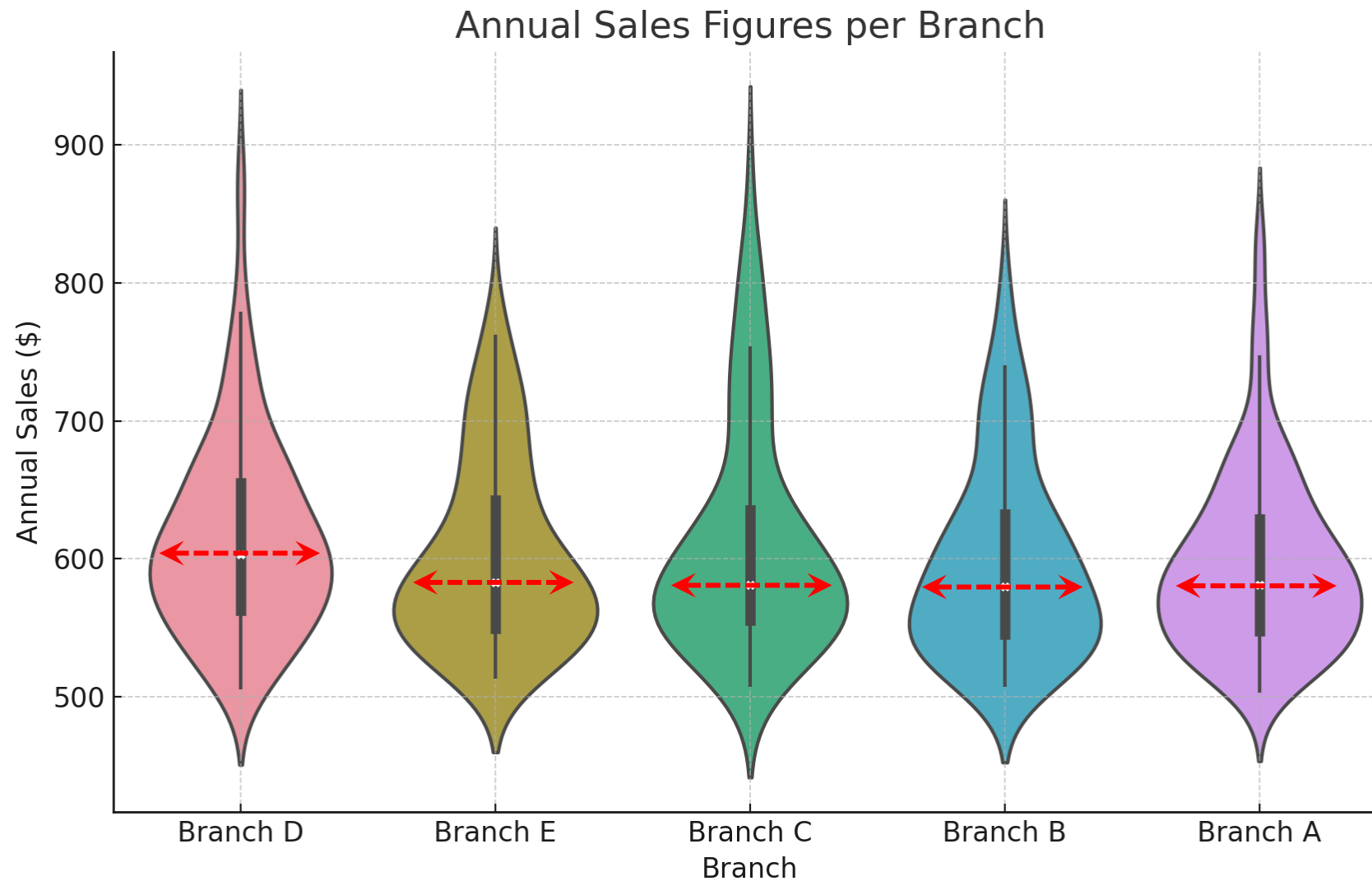
WHAT IS A VIOLIN PLOT?



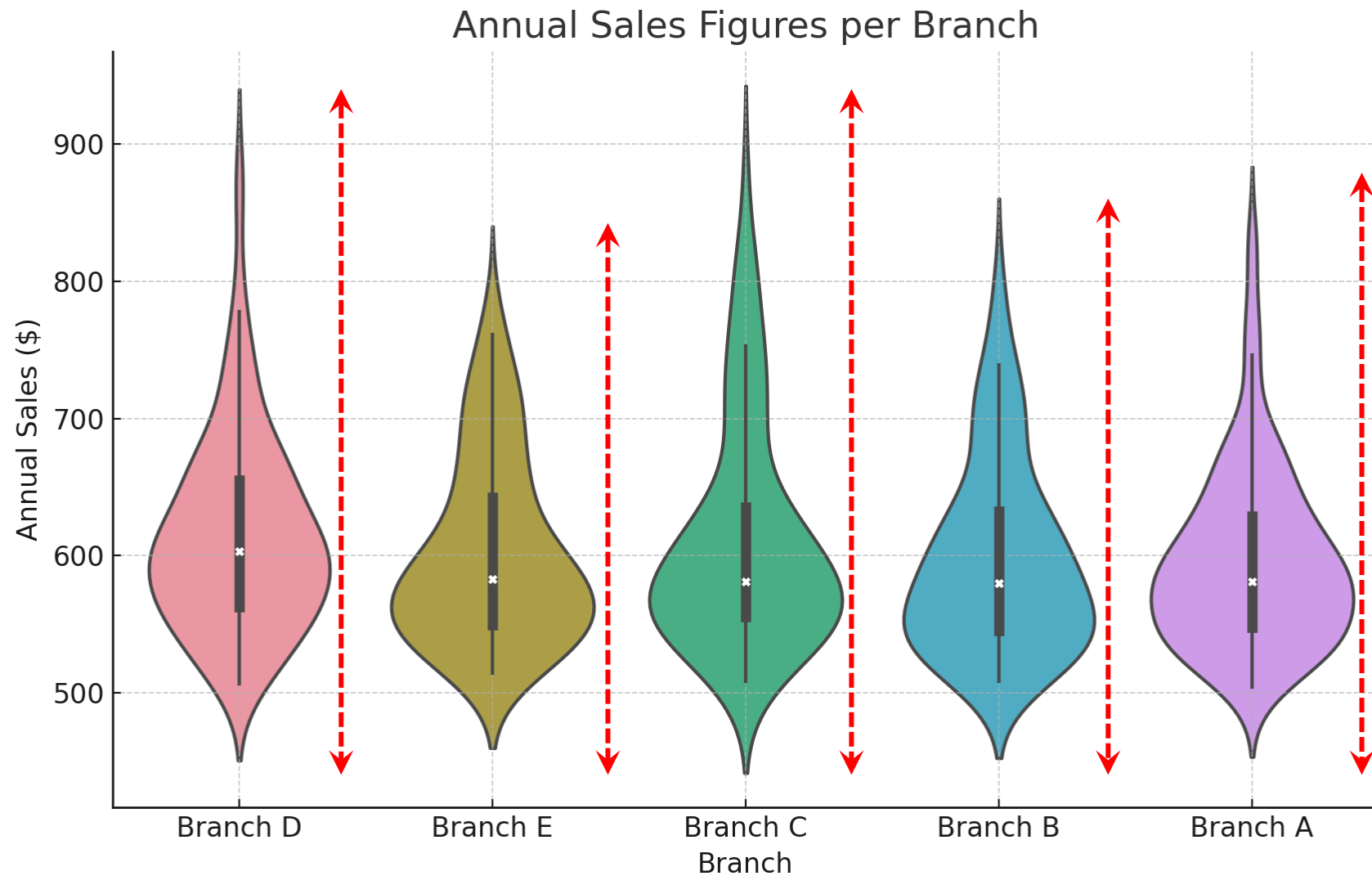
WHAT IS A VIOLIN PLOT?



WHAT IS A VIOLIN PLOT?



WHAT IS A VIOLIN PLOT?



CHAT GPT

WHY IS IT IMPORTANT?

WHY IS A VIOLIN PLOT IMPORTANT?



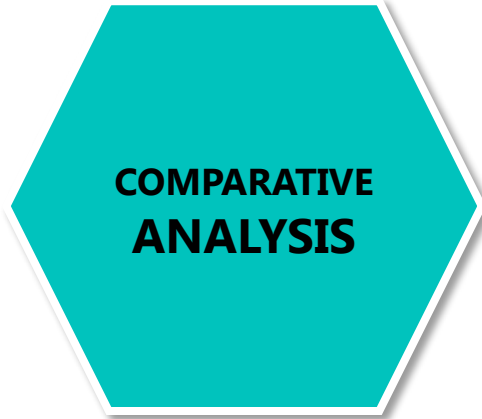
COMPREHENSIVE DATA DISTRIBUTION VISUALIZATION

- **DO NOT SHOW JUST SUMMARY STATISTICS**
- **SHOWS THE ENTIRE DISTRIBUTION OF DATA**



SPOTTING SUBTLETIES IN DATA

- **CAN REVEAL NUANCES BETTER THAN BOX PLOTS**
- **E.G. MULTIMODALITY (MULTIPLE PEAKS)**



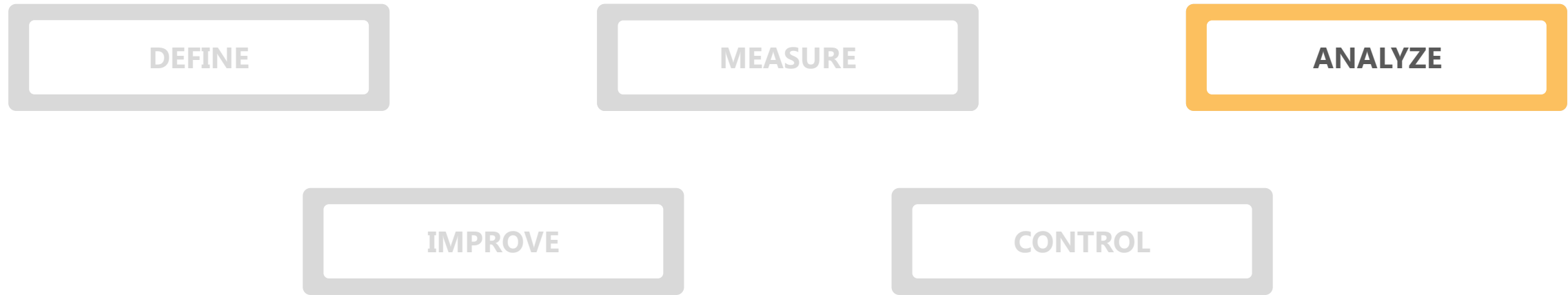
COMPARATIVE ANALYSIS

- **USEFUL TO COMPARE DISTRIBUTION OF DATA ACROSS SEVERAL GROUPS OR CATEGORIES SIDE-BY-SIDE**

The background image shows a person's hand typing on a laptop keyboard. Overlaid on the laptop screen is a semi-transparent interface for 'CHAT GPT'. The interface includes a refresh icon in the top left, the text 'CHAT GPT' in large letters, and two chat bubbles below it. The first bubble contains a person icon and three horizontal lines, while the second bubble contains a brain icon with gears. The entire scene is set against a dark background with abstract blue and purple light streaks and a grid of small glowing dots at the bottom.

HOW IS THE VIOLIN PLOT USED IN SIX SIGMA PROJECTS?

HOW IS THE VIOLIN PLOT USED IN SIX SIGMA PROJECTS?



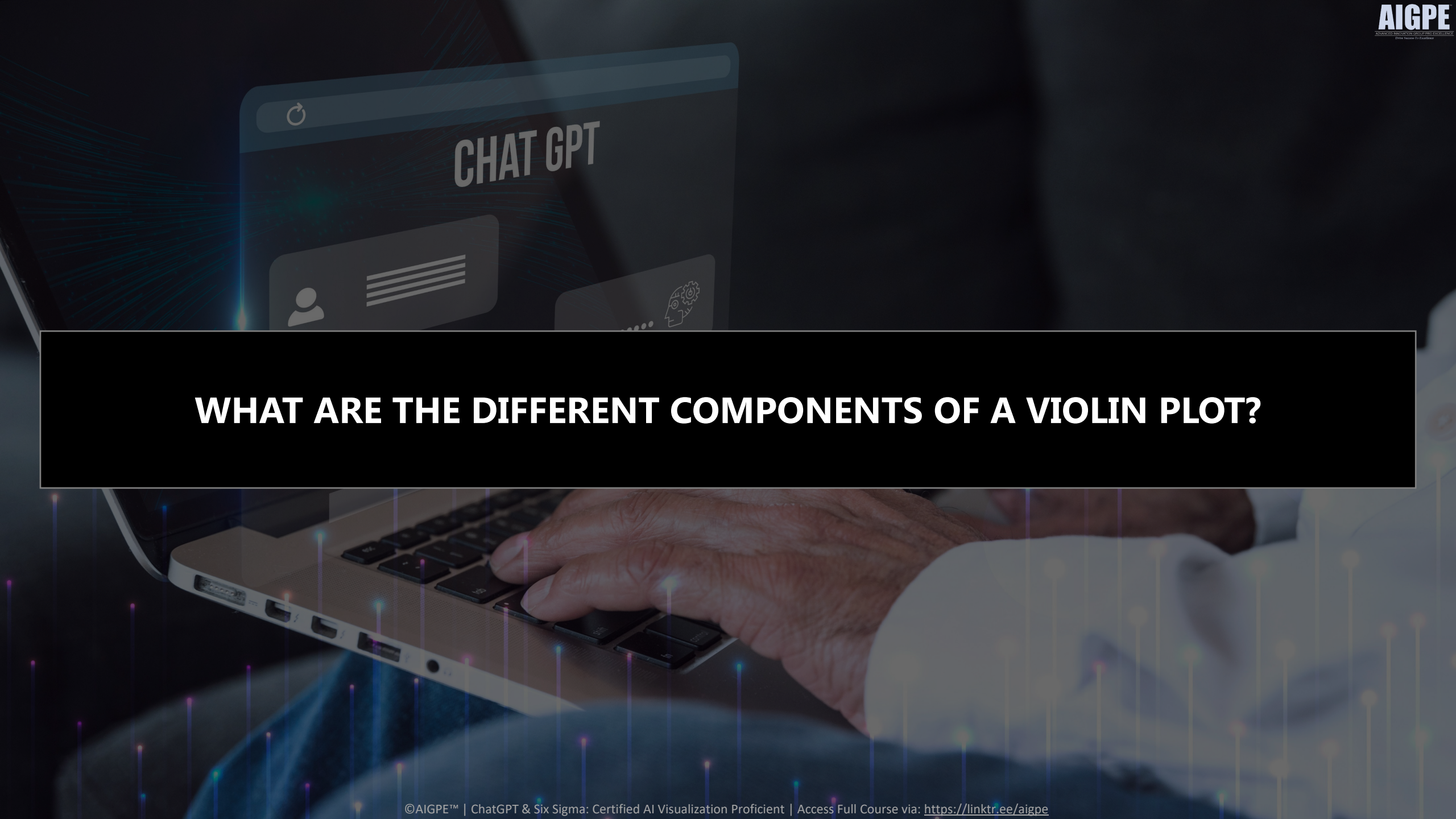
SHOWS THE DENSITY OF DATA AND REVEALS POTENTIAL MULTIMODAL DISTRIBUTIONS

INDICATE THE VARYING CAUSES OF VARIABILITY WITHIN A PROCESS

HELPS SIX SIGMA PROFESSIONALS TO IDENTIFY OUTLIERS AND PATTERS OF PROCESS BEHAVIOR

ENABLE TEAMS TO TARGET SPECIFIC ASPECTS OF THE PROCESS

HAS THE ABILITY TO CONCISELY PRESENT COMPLEX STATISTICAL INFORMATION

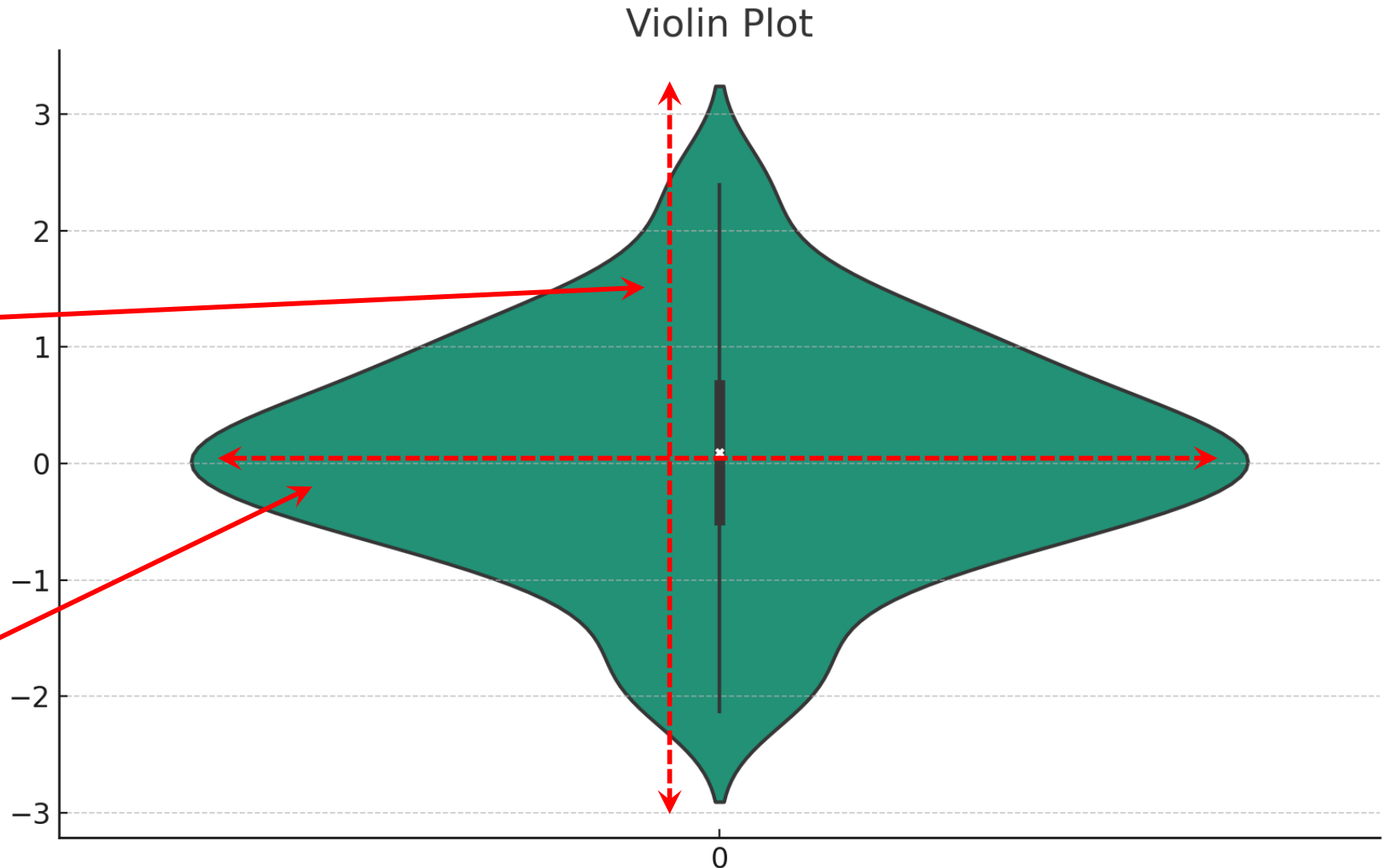
The background of the slide features a person's hand typing on a laptop keyboard. Overlaid on the top left is a semi-transparent interface for 'CHAT GPT', showing a refresh icon, the text 'CHAT GPT', and a chat bubble with a user icon and a response icon. The entire scene is set against a dark background with vertical light streaks and a network of nodes at the bottom.

WHAT ARE THE DIFFERENT COMPONENTS OF A VIOLIN PLOT?

BODY (VIOLIN SHAPE)

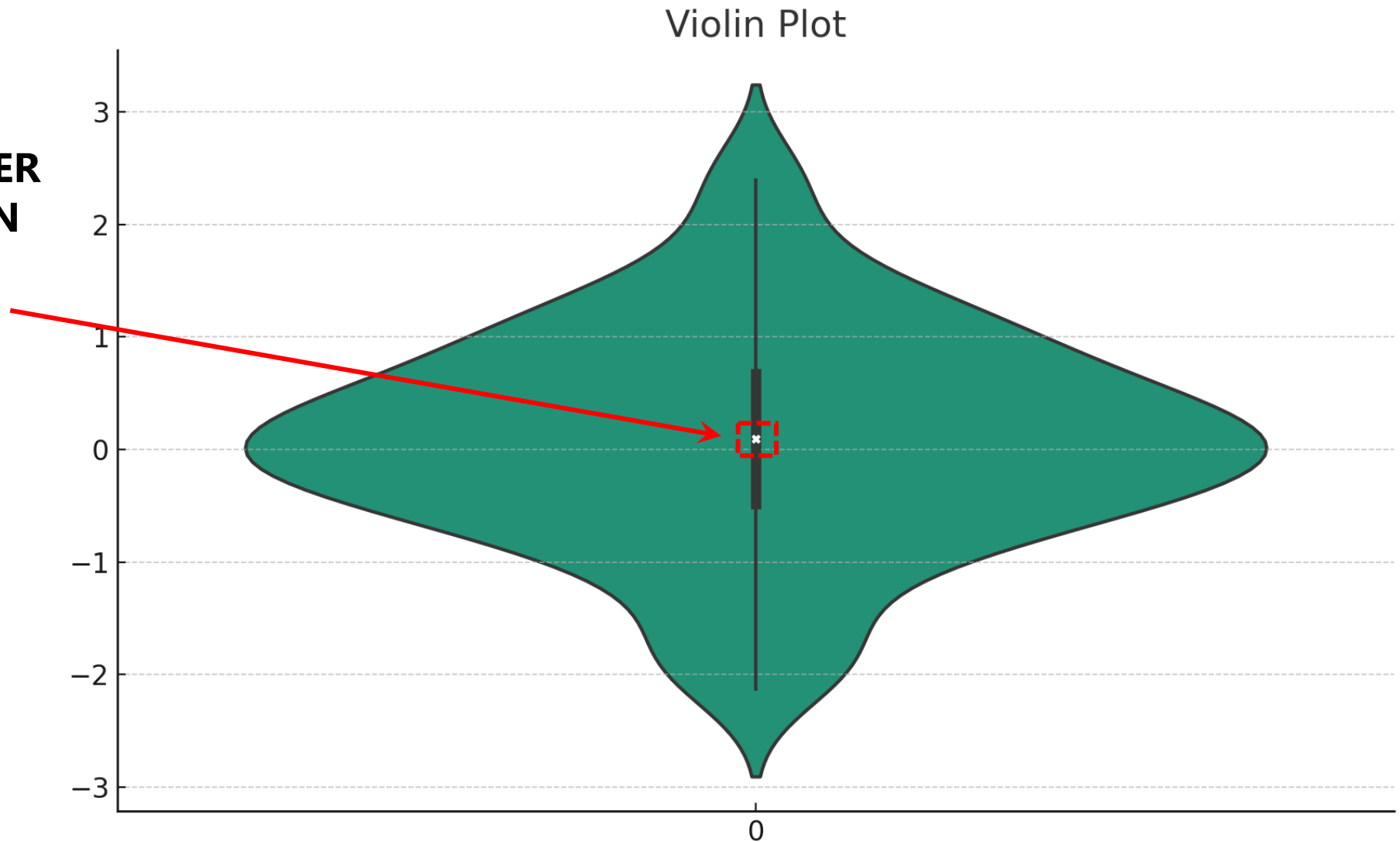
**WIDTH INDICATES THE
 FREQUENCY OF DATA
 POINTS AT DIFFERENT
 LEVELS**

**THE WIDE, CENTRAL
 PART DISPLAYS THE
 DENSITY OF THE DATA**



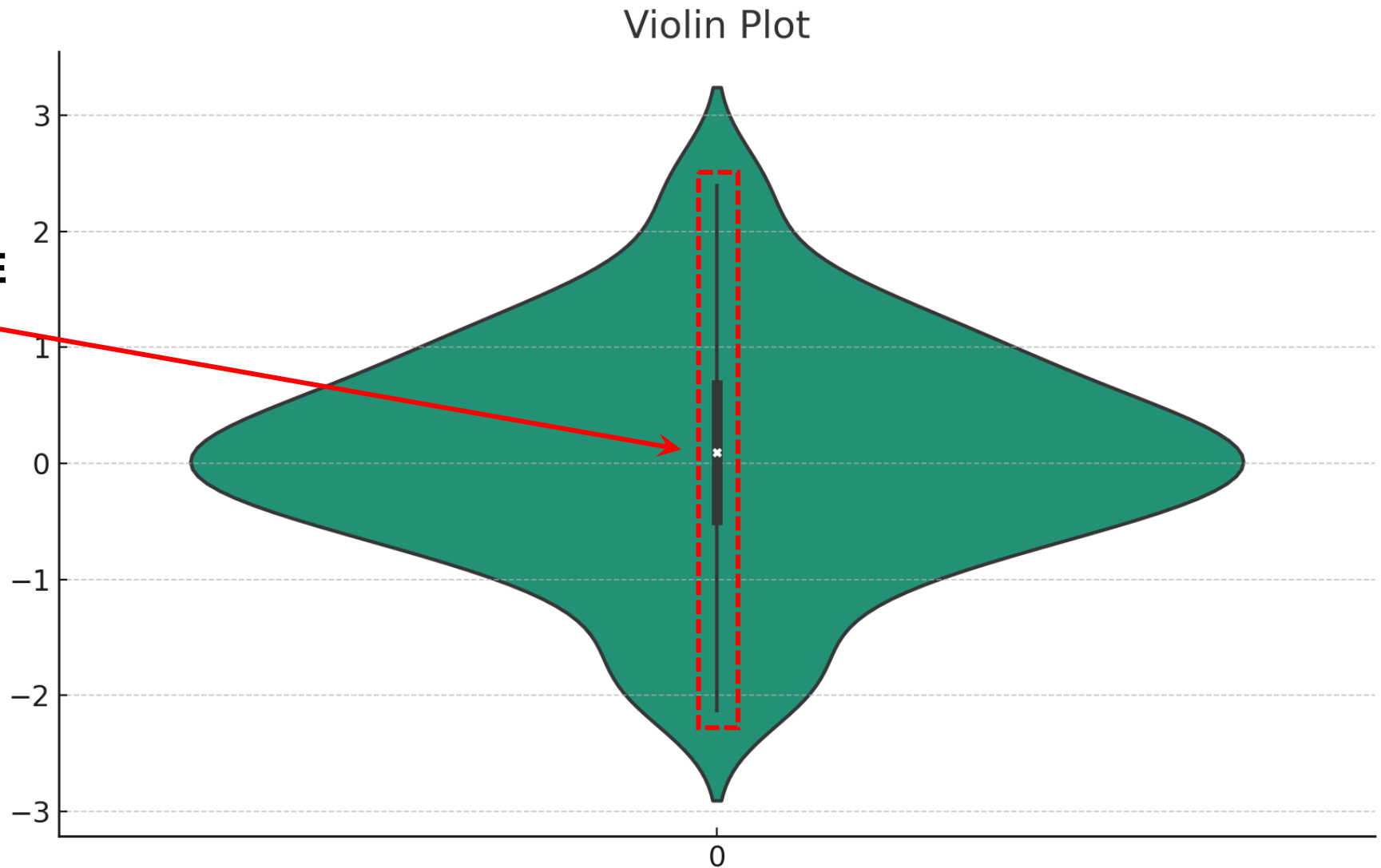
MEDIAN MARKER

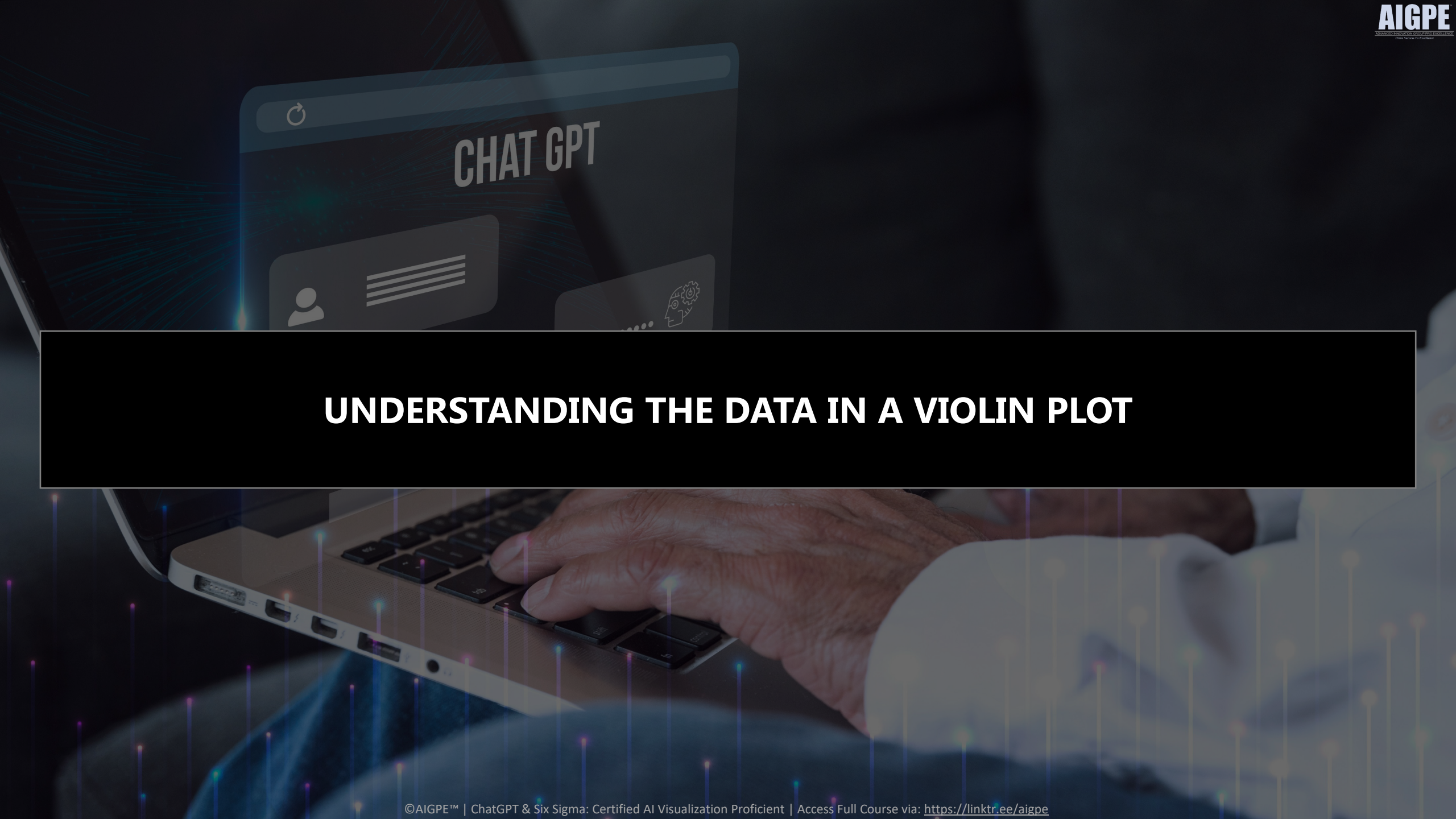
**THE MEDIAN MARKER
 SHOWS THE MEDIAN
 VALUE OF THE DATA**



BOX PLOT OVERLAY

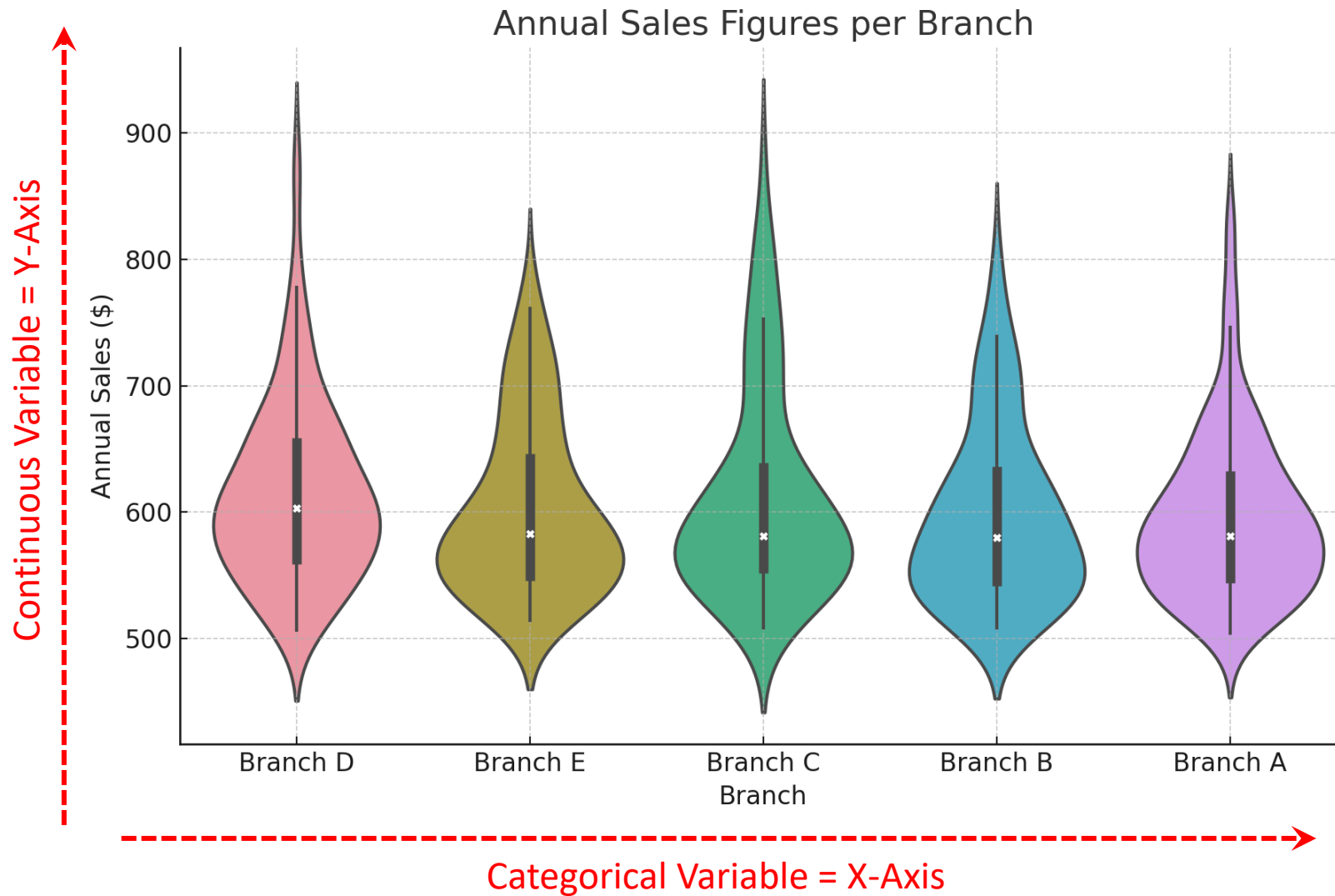
**THE MINI BOX PLOT
 INSIDE THE VIOLIN
 SHOWS THE
 INTERQUARTILE RANGE
 AND THE MEDIAN**





UNDERSTANDING THE DATA IN A VIOLIN PLOT

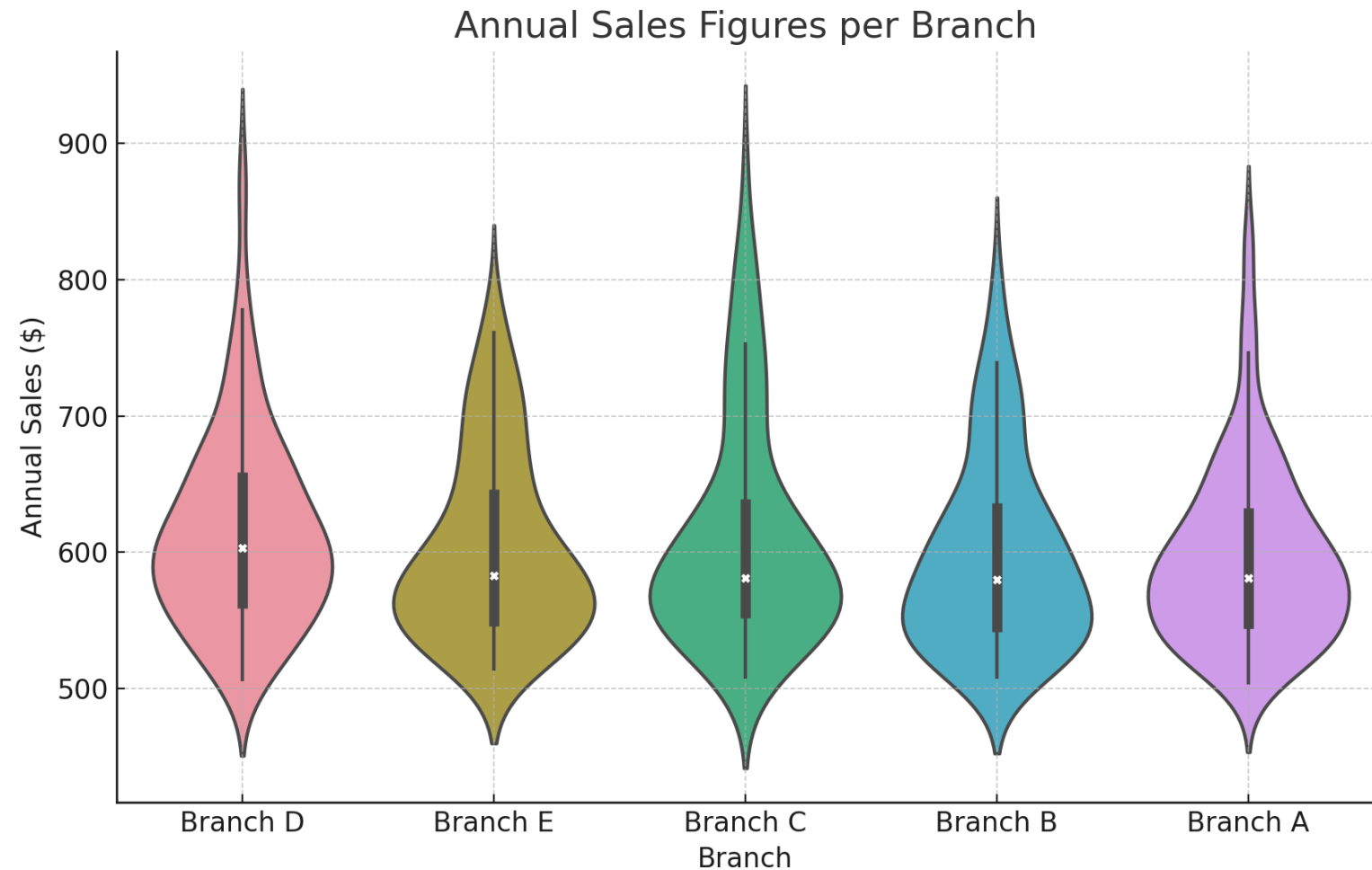
UNDERSTANDING THE DATA IN A VIOLIN PLOT



UNDERSTANDING THE DATA IN A VIOLIN PLOT

X-AXIS:

- TIME PERIODS
- GROUPS
- CATEGORIES

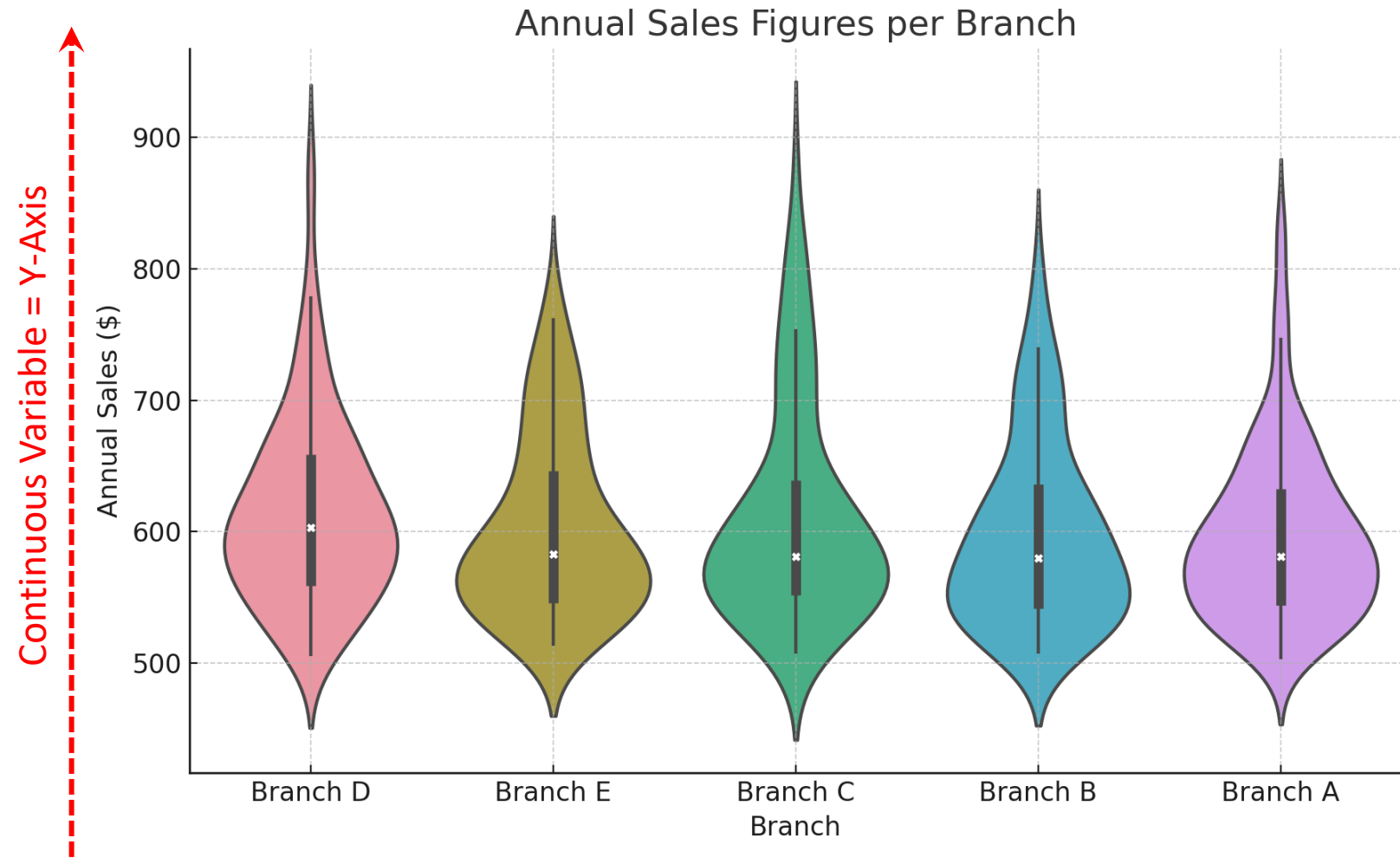


----->
Categorical Variable = X-Axis

UNDERSTANDING THE DATA IN A VIOLIN PLOT

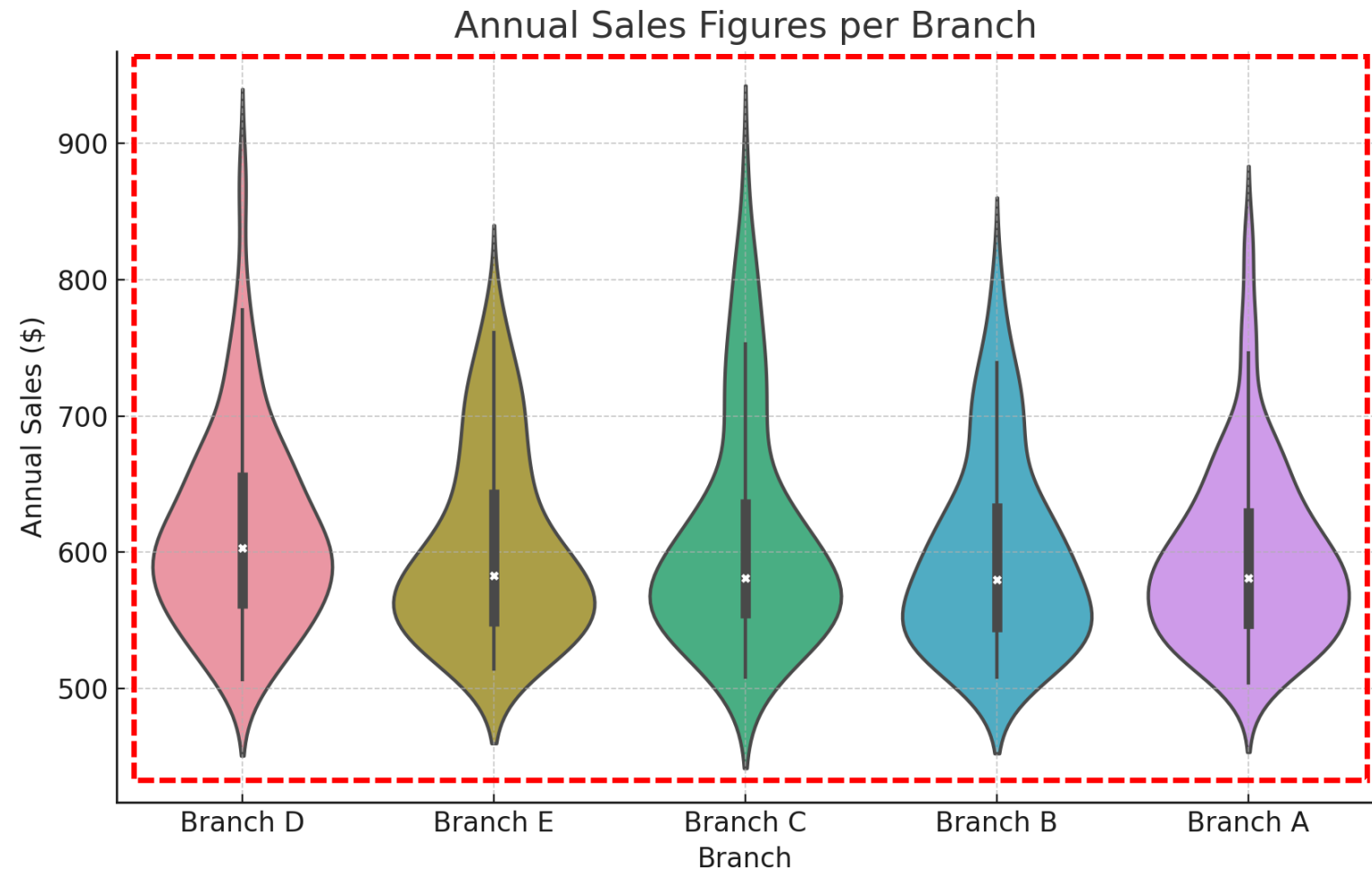
Y-AXIS:

- SCORES
- MEASUREMENTS
- COUNTS



UNDERSTANDING THE DATA IN A VIOLIN PLOT

THE PLOT VISUALLY REPRESENTS HOW THE DATA DISTRIBUTION VARIES ACROSS CATEGORIES, OFFERING INSIGHTS INTO DATA'S STRUCTURE & OUTLIERS



AIGPETM

ADVANCED INNOVATION GROUP PRO EXCELLENCE

Drive Success To Excellence